Toolformer: Language Models Can Teach Themselves to Use Tools

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Abstract

Language models (LMs) demonstrate remarkable abilities to solve new tasks from just a few examples or textual instructions. However, they struggle with basic tasks like arithmetic, factual lookup, and temporal reasoning. In this paper, we introduce Toolformer, a model trained to decide which tools to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction.

1. Self-Supervised Tool Use

Unlike existing approaches that rely on large-scale human annotation of tool usage, Toolformer is trained in a completely self-supervised manner: given a plain text dataset, the model itself samples candidate API calls, executes them, and keeps only those API calls that demonstrably reduce perplexity on predicting subsequent tokens.

2. Tools and Filtering Methodology
Toolformer integrates five diverse tools via textual API call syntax:
1. **Calculator:** Solves precise multi-step arithmetic expressions.
2. **Q&A; Retrieval System (Wikipedia Search):** Fetches up-to-date factual snippets.
3. **Calendar / Clock:** Provides current date, time, and temporal anchoring.
4. **Machine Translation System:** Translates non-English queries.
5. **Machine Wiki Search:** Looks up structured entities.

Loss Filtering Criterion: An API call c is inserted into sequence x. Let L_without be the cross-entropy loss of predicting future tokens without the tool call, and L_with be the loss with the tool response inserted. The API call is retained if and only if: *L_with - L_without* < *-delta* where delta > 0 is a filtering threshold.